

PILLA SRI SAI RAHUL

MECHATRONICS ENGINEERING | EMBEDDED SYSTEMS | SOFTWARE & AI

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PROFILE

First-semester Mechatronics Engineering student at Mahindra University building independent projects across software, embedded systems, and AI-assisted applications. Project work includes audio-model inference experiments, memory optimization, voice-based electronics troubleshooting, and developer-support tooling. Interested in opportunities across mechatronics, embedded/IoT, and software engineering.

EDUCATION

Mahindra University — B.Tech, Mechatronics Engineering | 2026–2030 · First semester

Class XII — Andhra Pradesh State Board — 89% | 2026

Class X — CBSE — 85.4% | 2024

SELECTED PROJECTS

VI-071 — Audio-to-Sheet-Music Application

github.com/1RY-O/Vi-071

- Developing an audio-to-sheet-music workflow using the MuScriptor transcription model, a web application, and a separate Python inference worker.
- Tested inference-memory optimizations including KV-cache adjustments and FlashAttention trials; observed memory use decrease from an earlier ~1.15 GB baseline run to ~690 MB in an optimized run.
- Evaluated ONNX inference and INT8 quantization; did not adopt quantization after testing showed an unacceptable accuracy trade-off. Used Cloudflare Tunnel to connect to the development worker.

Bobby — Developer Troubleshooting Assistant

bobby-neon.vercel.app • github.com/1RY-O/Bobby

- Independently built a developer troubleshooting assistant that organizes investigation into planning, error/code examination, and remediation stages.
- Built with Next.js, FastAPI, LangGraph, React Flow, and Tailwind CSS; public repository includes local run and test instructions.

CircuitMate — Voice-Native Electronics Troubleshooting Assistant

circuitmate-chi.vercel.app • circuitmate.onrender.com

- Independently developed a voice-oriented assistant for electronics and embedded-system troubleshooting.
- Integrated AssemblyAI voice-agent functionality with a backend and hardware-focused knowledge base for hands-busy debugging workflows involving Arduino and ESP32.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, HTML, CSS

Embedded: Arduino, ESP32, sensors, actuators

Web & backend: React, Vite, Node.js, Express, Next.js, FastAPI, REST APIs

AI & development: Gemini API, Claude, AssemblyAI, LangGraph, structured outputs, prompt engineering, OpenCode, Cline

Tools & platforms: Git, GitHub, VS Code, Fedora/Linux, Tinkercad, Firebase, Firestore, Vercel, Render

CERTIFICATION

BE10X — AI Tools & Claude Workshop | Certificate of Completion | August 2026

ACTIVITIES

Member, Blockchain Club and Computer Science Club (Enigma), Mahindra University.